AIRPLANE CRASHES & FATALITIES – SAFETY ANALYSIS

Interactive Excel Dashboard Project

INTRODUCTION

- This project analyzes historical airplane crash data from 1908 to 2025.
- The focus is on identifying trends, patterns, and safety insights over time.
- Key metrics such as total crashes, fatalities, average fatalities, and fatality rate are calculated.
- An interactive Excel dashboard is developed to visualize trends and support decision-making.
- The analysis aims to highlight high-risk aircraft, operators, and improve aviation safety awareness.

Objective:

- Analyze airplane crash data to identify trends and safety insights
- Evaluate the impact on passengers (Fatalities & Crashes)
- Build an interactive dashboard to support decision-making

Business Problem:

- Aviation safety analysis over decades
- Identify high-risk aircraft, operators, and trends

index	Decade	Year	Date	Time	Location	Operator	Flight #	Route	Departure	Destination	Type	Registration	cn/In	Aboard	Fatalities	Ground	Summary	Column1	Column2	Co
0	1900	1900	09/17/1908	17:18:00	Fort Myer, V	Military - U.S. Army		Demonstra	Null	Null	Wright Flye	Null	1	2.00	1.00	0.00	During a demonstration flight, a U.S. Army			
1	1900	1900	07-12-1912	06:30:00	AtlantiCity, I	Military - U.S. Navy		Test flight	Null	Null	Dirigible	Null		5.00	5.00	0.00	First U.S. dirigible Akron exploded just offs			
2	1900	1900	08-06-1913		Victoria, Bri	Private	-		Null	Null	Curtiss sea	Null		1.00	1.00	0.00	The first fatal airplane accident in Canada			
3	1900	1900	09-09-1913	18:30:00	Over the No	Military - German Navy			Null	Null	Zeppelin L-	Null		20.00	14.00	0.00	The airship flew into a thunderstorm and e			
4	1900	1900	10/17/1913	10:30:00	Near Johan	Military - German Navy			Null	Null	Zeppelin L-	Null		30.00	30.00	0.00	Hydrogen gas which was being vented was			
5	1900	1900	03-05-1915	01:00:00	Tienen, Belg	Military - German Navy			Null	Null	Zeppelin L-	Null		41.00	21.00	0.00	Crashed into trees while attempting to lan			
6	1900	1900	09-03-1915	15:20:00	Off Cuxhave	Military - German Navy			Null	Null	Zeppelin L-	Null		19.00	19.00	0.00	Exploded and burned near Neuwerk Island			
7	1900	1900	07/28/1916		Near Jambo	Military - German Army			Null	Null	Schutte-La	Null		20.00	20.00	0.00	Crashed near the Black Sea, cause unkno			
8	1900	1900	09/24/1916	01:00:00	Billericay, E	Military - German Navy			Null	Null	Zeppelin L-	Null		22.00	22.00	0.00	Shot down by British aircraft crashing in fla			
9	1900	1900	10-01-1916	23:45:00	Potters Bar,	Military - German Navy			Null	Null	Zeppelin L-	Null		19.00	19.00	0.00	Shot down in flames by the British 39th Ho			
10	1900	1900	11/21/1916		Mainz, Gern	Military - German Army			Null	Null	Super Zep	Null		28.00	27.00	0.00	Crashed in a storm.			
11	1900	1900	11/28/1916	23:45:00	Off West Ha	Military - German Navy			Null	Null	Zeppelin L-	Null		20.00	20.00	0.00	Shot down by British anti-aircraft fire and a			
12	1900	1900	03-04-1917		Near Gent, I	Military - German Army			Null	Null	Airship	Null		20.00	20.00	0.00	Caught fire and crashed.			
13	1900	1900	03/30/1917		Off Northern	Military - German Navy			Null	Null	Schutte-La	Null		23.00	23.00	0.00	Struck by lightning and crashed into the Ba			
14	1900	1900	05/14/1917	05:15:00	Near Texel I	Military - German Navy			Null	Null	Zeppelin L-	Null		21.00	21.00	0.00	Crashed into the sea from an altitude of 3,			
15	1900	1900	06/14/1917	08:45:00	Off Vlieland	Military - German Navy			Null	Null	Zeppelin L-	Null		24.00	24.00	0.00	Shot down by British aircraft.			
16	1900	1900	08/21/1917	07:00:00	Off western	Military - German Navy			Null	Null	Zeppelin L-	Null		18.00	18.00	0.00	Shot down by British aircraft.			
17	1900	1900	10/20/1917	07:45:00	Near Lunevi	Military - German Navy			Null	Null	Zeppelin L-	Null		18.00	18.00	0.00	Shot down by French anti-aircraft fire.			
18	1900	1900	04-07-1918	21:30:00	Over the Me	Military - German Navy			Null	Null	Zeppelin L-	Null		23.00	23.00	0.00	Exploded and crashed into the sea off the s			
19	1900	1900	05-10-1918		Off Helgolar	Military - German Navy			Null	Null	Zeppelin L-	Null		22.00	22.00	0.00	Shot down by British aircraft crashing from			
20	1900	1900	08-11-1918	10:00:00	Ameland Isl	Military - German Navy			Null	Null	Zeppelin L-	Null		19.00	19.00	0.00	Shot down by british aircraft.			
21	1900	1900	12/16/1918		Elizabeth, N	US Aerial Mail Service			Null	Null	De Havillar	97		1.00	1.00	0.00				
22	1900	1900	05/25/1919		Cleveland, (	US Aerial Mail Service			Null	Null	De Havillar	61		1.00	1.00	0.00	Caught fire in midair. The pilot leaped from			
23	1900	1900	07/19/1919		Dix Run, Per	US Aerial Mail Service			Null	Null	De Havillar	82		1.00	1.00	0.00				
24	1900	1900	10-02-1919		Newcastle,	Aircraft Transport and Travel			Null	Null	De Havillar	Null		1.00	1.00	0.00				
25	1900	1900	10/14/1919		Cantonsville	US Aerial Mail Service			Null	Null	Curtiss R-4	32		1.00	1.00	0.00				
26	1900	1900	10/20/1919		English Cha	Aircraft Transport and Travel			Null	Null	De Havillar	G-EAHG		13.00	9.00	0.00				
27	1900	1900	10/30/1919		Long Valley,	US Aerial Mail Service			Null	Null	De Havillar	77		1.00	1.00	0.00				
28	1900	1900	03-10-1920		New Paris, I	US Aerial Mail Service			Null	Null	De Havillar	65		1.00	1.00	0.00				
29	1900	1900	03/30/1920		Newark, Ne	US Aerial Mail Service			Null	Null	De Havillar	72		1.00	1.00	0.00				
30	1900	1900	04-10-1920		Newark, Ne	US Aerial Mail Service			Null	Null	Curtiss R-4	49		2.00	1.00	0.00				
31	1900	1900	04-11-1920		Heller Field,	US Aerial Mail Service			Null	Null	Curtiss JN-	44		2.00	1.00	0.00				
32	1900	1900	05-12-1920		Oskaloosa,	US Aerial Mail Service			Null	Null	De Havillar	79		2.00	1.00	0.00				
33	1900	1900	06-06-1920		Cleveland, (	US Aerial Mail Service			Null	Null	De Havillar	204		1.00	1.00	0.00				
34	1900	1900	08/16/1920		Bedford, En	By Air			Null	Null	Armstrong	G-EALW		1.00	1.00	0.00				
35	1900	1900	08/16/1920		College Parl	US Aerial Mail Service			Null	Null	De Havillar	317		1.00	1.00	0.00				
36	1900	1900	09-01-1920		Morristown,	US Aerial Mail Service			Null	Null	Junkers JL-	305		2.00	2.00	0.00				
37	1900	1900	09/15/1920		Pemberville	US Aerial Mail Service			Null	Null	Junkers F-	308		2.00	2.00	0.00				
38	1900	1900	09/27/1920		Hillersburg,	US Aerial Mail Service			Null	Null	De Havillar	316		1.00	1.00	0.00				
39	1900	1900	10-02-1920		Off Port Ven	Latecoere Airlines			Null	Null	Salmson 2	F-ALAI	31	2.00	2.00	0.00				
40	1900	1900	10-05-1920		Valencia, Sp	Latecoere Airlines			Null	Null	Breguet 14	F-ALTA	20	2.00	2.00	0.00	Crashed while landing.			

DATASET: AIRPLANE CRASHES AND FATALITIES (CSV)
NUMBER OF RECORDS: 2000+ CRASH RECORDS
COLUMNS: DATE, OPERATOR, AIRCRAFT TYPE, FATALITIES, ABOARD, LOCATION
TIME RANGE: 1908–2025

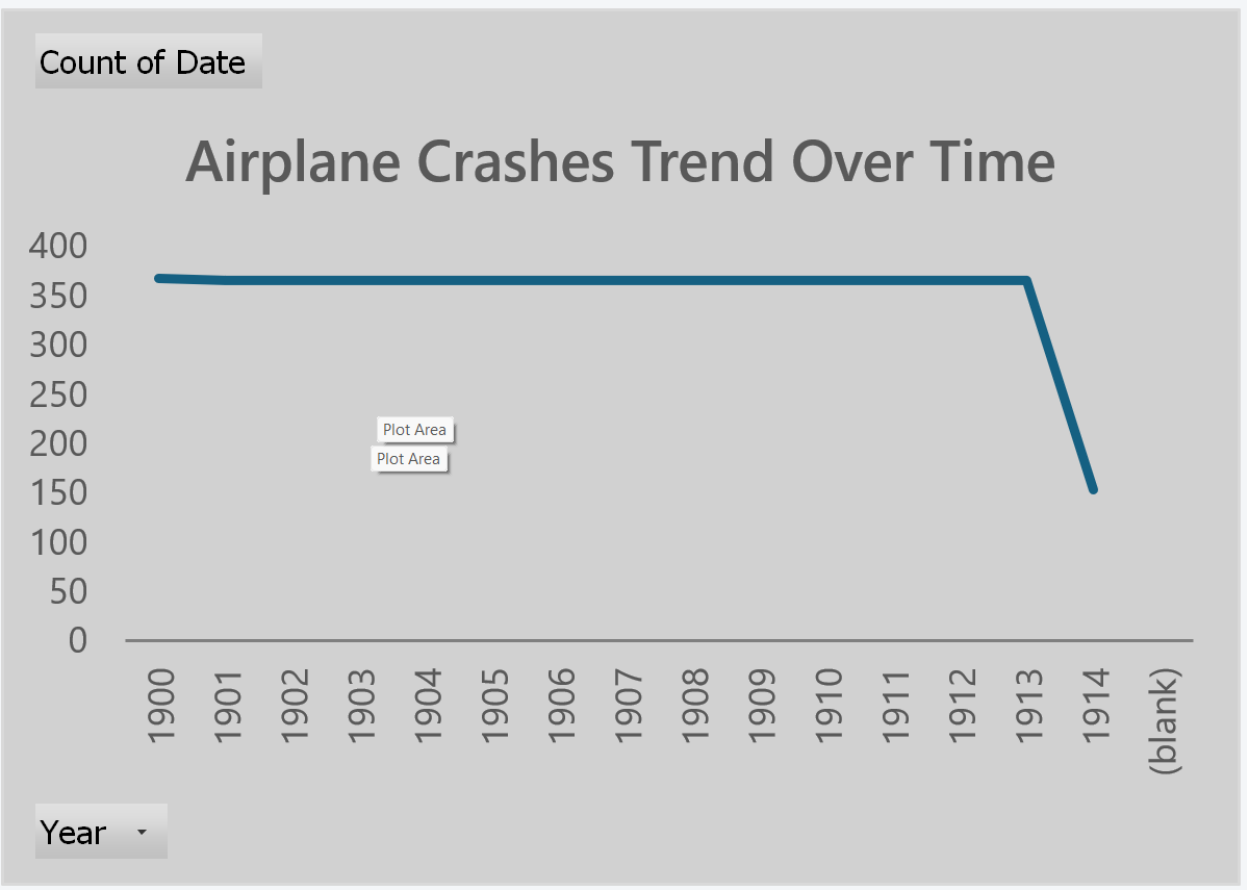


CHART DESCRIPTION

Key Insights

Airplane crash counts remain largely stable from 1900–1913.

No clear upward or downward trend during most of the period.

A sharp drop in the final year suggests a possible data gap or external factor.

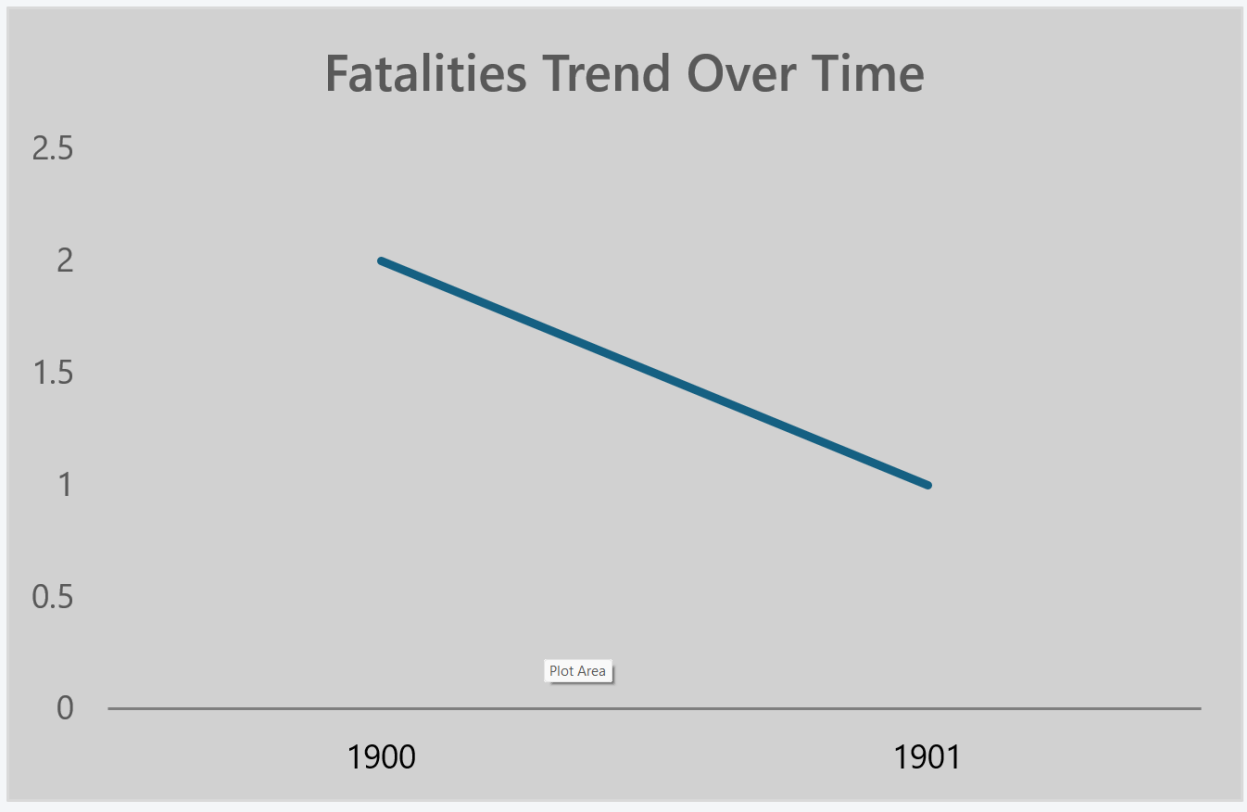


CHART DESCRIPTION

Key Insights

Fatalities show a declining trend over the observed period.

Average fatalities drop from ~2.0 to ~1.0, indicating improved safety or fewer severe incidents.

Overall severity of crashes appears to be reducing over time.

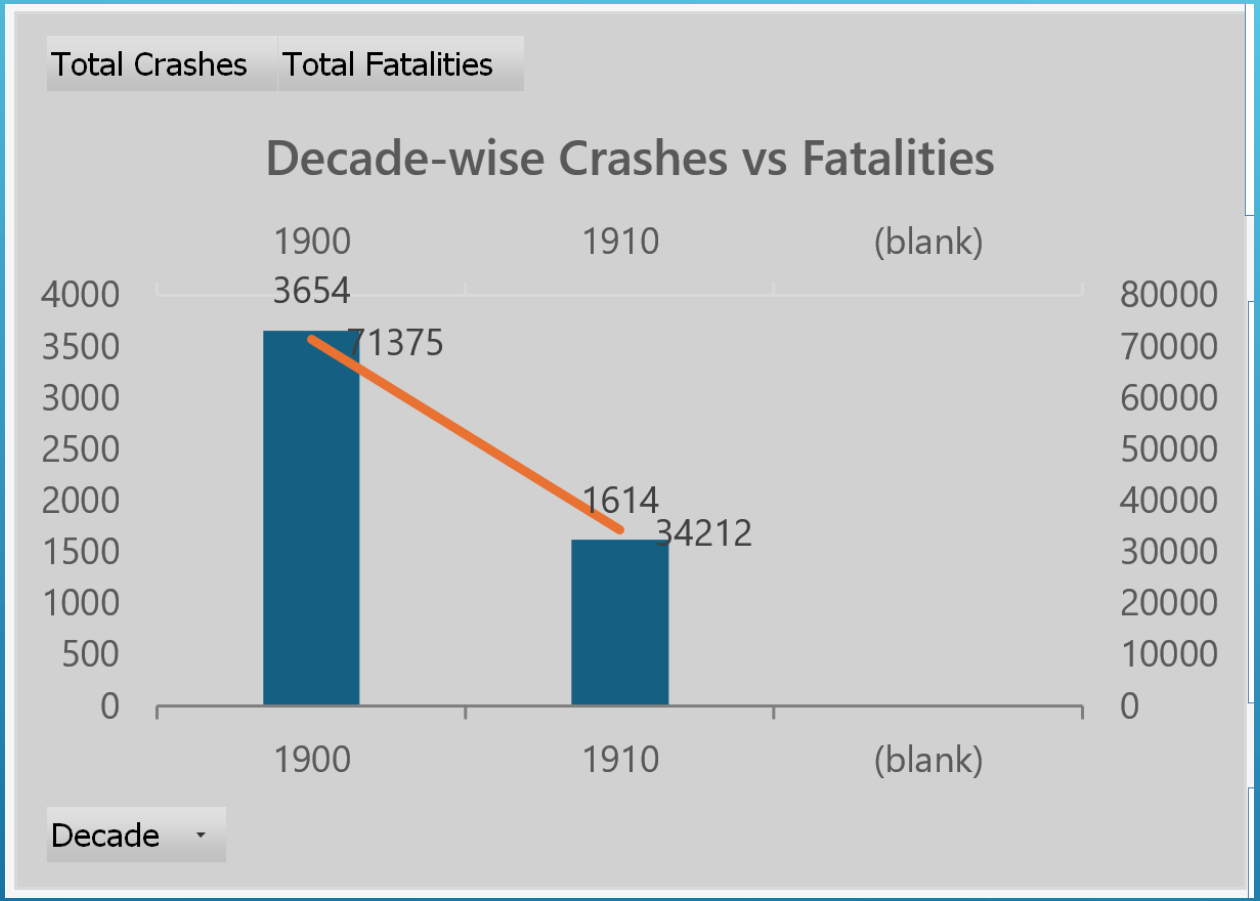


CHART DESCRIPTION

Key Insights

1900s decade records the highest crashes and fatalities, indicating early aviation risks.

Both total crashes and fatalities decline in the 1910s, showing safety improvements over time.

Fatalities drop more sharply than crashes, suggesting reduced crash severity.

CHART DESCRIPTION

Key Insights

Fatalities are highly concentrated among a small number of operators.

A few operators account for a disproportionately large share of total fatalities.

Indicates uneven safety performance across operators.

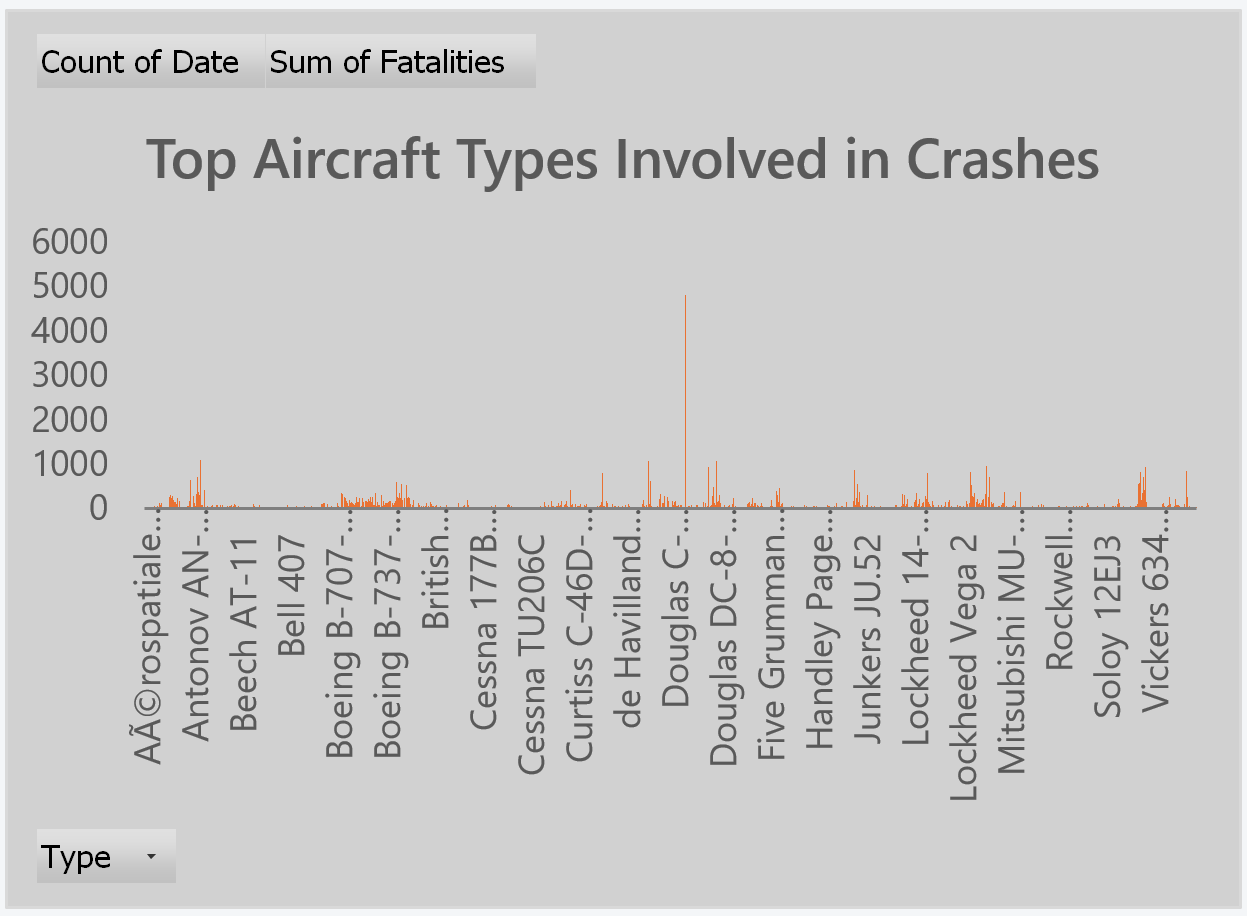


CHART DESCRIPTION

Key Insights

Crash involvement is spread across many aircraft types, with a few models standing out.

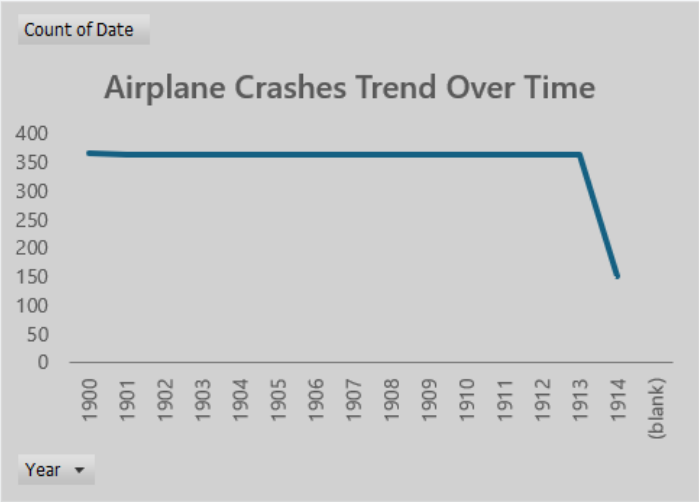
Certain aircraft (e.g., widely used or long-serving models) show higher crash counts, likely due to greater exposure.

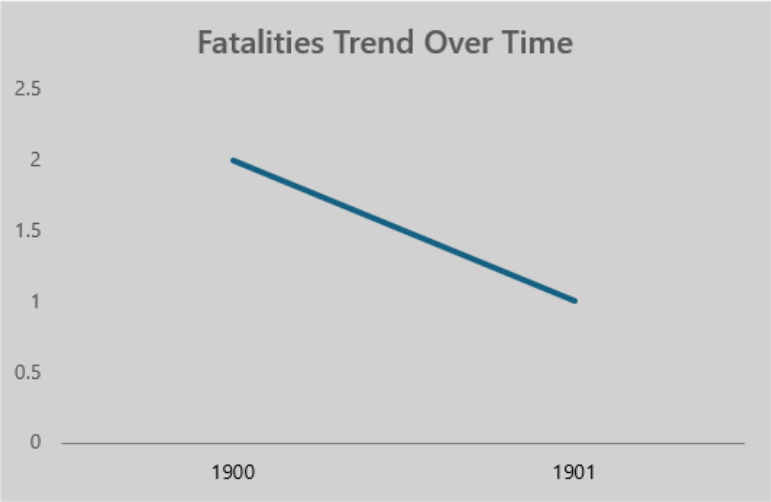
Indicates that usage volume and operational lifespan strongly influence crash frequency, not just design.

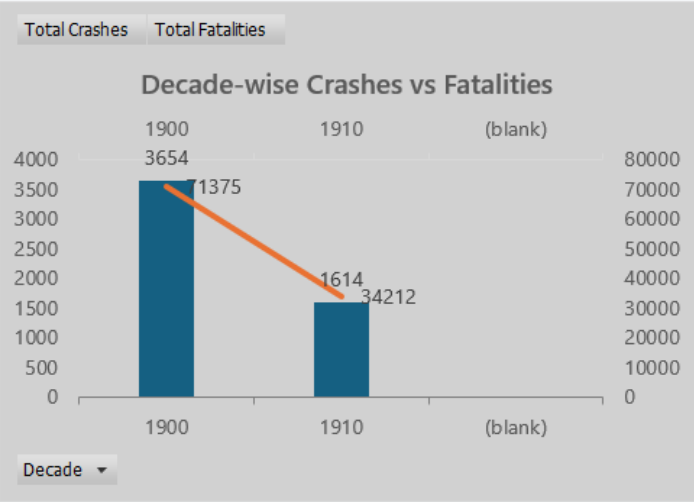
DASHBOARD

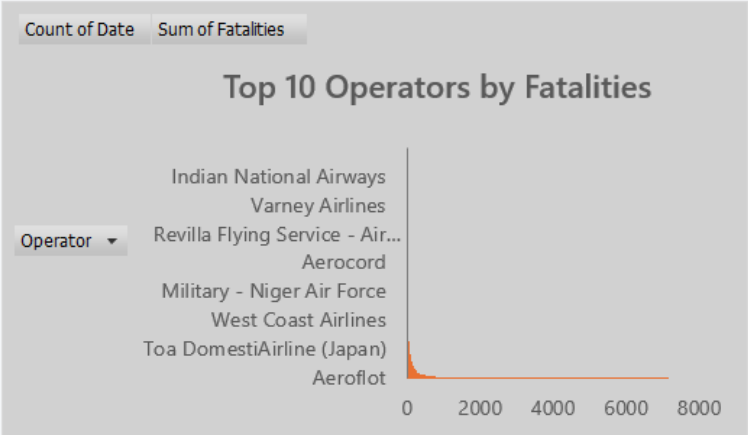
Airplane Crashes & Fatalities – Safety Analysis Dashboard

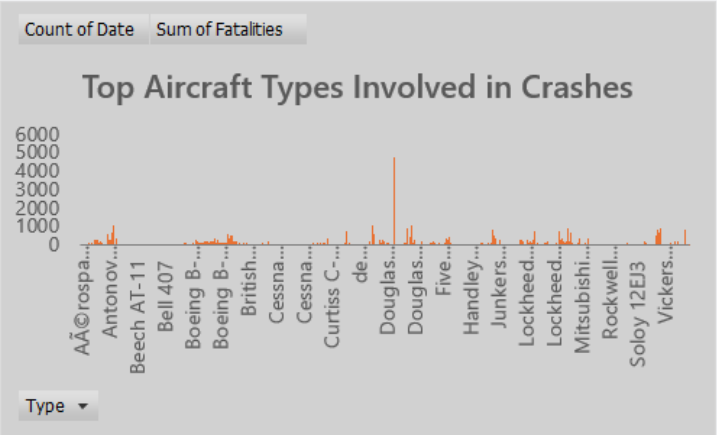
Total Crashes	Total Fatalities	Avg Fatalities	Fatality Rate(%)
5268	105587.00	20.04	72.90%











Year

1910 | 1911 | 1912 | 1913 | 1914

Operator

A B Aerotransport
AÃ©reo Ruta Maya
AÃ©rotaxi Cachanilla

Type

AÃ©rospatiale SE-210 C...
AÃ©rospatiale/Aeritalia ...
AAC-1 Toucan

Decade

1900
1910

CONCLUSION

- The interactive dashboard enables detailed trend analysis of airplane crashes and fatalities over decades.
- Key KPIs and visual charts provide quick, actionable insights for aviation safety monitoring.
- Historical analysis shows a decline in fatality rates due to improved safety measures.
- Dashboard highlights high-risk aircraft types and operators for targeted safety interventions.
- The tool can support data-driven decision-making to enhance aviation safety policies and practices.

THANK YOU